

KEY

Science 24

UNIT C WORKBOOK



SENIOR HIGH SCHOOL

Fort Saskatchewan 780 992 0101

Sherwood Park 780 464 1899

Vegreville 780 632 7998



Chapter 9: Disease

Define the following:

Pasteurization

process in which a food is heated to kill the microbes that make it spoil (p. 158)

Disease

illness; something that prevents the normal functioning of the body (p. 158)

Microbes

micro-organisms (p. 158)

Pathogens

disease-causing agents such as bacteria, viruses, fungi, and protists (p. 159)

Bacteria

microscopic, one-celled organisms; may be beneficial, harmful, or neither (p. 159)

Viruses

microscopic, non-cellular disease-causing agents; see also *pathogens* (p. 159)

Fungi

organisms that live off dead and decaying organisms or are parasitic on living organisms; see also *parasites* (singular = fungus) (p. 162)

Protist

single-celled organisms that share some characteristics with both animals and plants; some cause serious disease, such as malaria; see also *pathogens* (p. 162)

Parasites

living organisms that get their nourishment from a host organism; cannot live independently (p. 162)

KEY

Communicable Disease

Illnesses that can be spread from one organism to another (p. 163)

Non-communicable Disease

illnesses that are not caused by pathogens and so cannot be spread from one organism to another (p. 163)

Botulism

form of food poisoning caused by bacteria (p. 163)

Salmonellosis

form of food poisoning caused by bacteria (p. 163)

Sanitary

maintaining clean, germfree conditions to help prevent disease (p. 164)

Poverty

circumstance in which basic needs for food, water, health care, and education are not met due to economic conditions (p. 166)

Staph bacteria

staphylococcus; cause of food poisoning (p. 168)

Sterilization

process that uses steam to destroy microscopic organisms such as bacteria (p. 170)

Aseptic

free of pathogens (p. 170)

KEY

1. Describe 5 factors that may interact with one another and possibly make you sick.
(pg. 158)

- **environment** — A poor environment may cause disease.
For example, air pollution can damage lungs.
People in polluted environments may suffer from lung disease.
- **genetics** — People can inherit diseases from their birth parents.
For example, some people suffer from muscular dystrophy.
- **diet** — A poor diet may result in an illness. For example, a diet high in fats may cause heart disease; a diet low in calcium may lead to osteoporosis.
- **lifestyle** — An unhealthy lifestyle may result in disease.
For example, drug or alcohol abuse may lead to liver problems; careless sexual practices may cause STDs.
- **pathogens** — Pathogens such as bacteria, viruses, fungi, and protists cause diseases. Athlete's foot is an example of a fungal disease.
- **stress** — Someone who is stressed may not eat properly or exercise regularly.
As a result, stress may result in high blood pressure and/or heart disease.

2. What did the following scientists discover.

a. Robert Koch (pg. 158)

**Robert Koch identified the pathogen that causes anthrax.
He discovered that a pathogen can spread from sick person to healthy person.
He also discovered that hot steam was effective in killing pathogens.
He used steam to destroy living organisms. This process is called sterilization.**

b. Louis Pasteur (pg. 158)

**Louis Pasteur discovered that a microbe was causing the spoilage of wine.
He heated grape juice to kill the microbes.
This technique is still used today and is called pasteurization.**

c. Florence Nightingale (pg. 170)

reduced the death rate by making sure the military hospital and patients were kept clean, improving water quality, and bandaging wounds to prevent infections from spreading to other patients.

d. Joseph Lister (pg. 170)

**realized that many patients died of infections following surgery.
He began spraying carbolic acid on patient wounds and into the air of operating theatres. Carbolic acid kills pathogens on surfaces and on skin.**

KEY

3. List the conditions that bacteria need to grow. Explain why each is important. (pg. 159)

Temperature — Most bacteria are active at warm temperatures.

Refrigeration or freezing slows bacterial growth.

Bacteria die when temperatures get too high.

Moisture — Bacteria need water. Many cannot reproduce without it.

Food — Bacteria thrive on sugar, protein, or fats.

4. What is a pathogen? Give three examples of pathogens. (pgs. 159 to 162)

Pathogens are microbes that cause disease.

examples: bacteria, viruses, fungi, protists.

5. Explain how viruses such as the AIDS virus spread. (pg. 159)

Viruses spread by taking over cells in the body.

They use the part of the body cell usually used by body cells to reproduce themselves.

Viruses reproduce quickly and take over other body cells.

6. How do protists spread? (pg. 162)

Protists such as Giardia spread when someone drinks contaminated water or touches an infected person's hands.

Protists such as head lice spread by contact.

7. What causes beaver fever? (pg. 162)

Giardia is the most common protist in the world.

In Canada, it is also called beaver fever because beavers often carry and spread the protist.

Giardia lives in the intestines of infected humans or animals.

Giardia may be found in water, food, soil, or surfaces that have been contaminated with the feces from infected humans or animals.

You can become infected after you drink the contaminated water.

8. What causes athlete's foot? (pg. 162)

Athlete's foot is caused by a fungus that lives on the skin.

It breeds under warm, moist conditions such as those found in sweaty socks and shoes.

KEY

9. What might happen to the bacteria in a hamburger you left on the counter for a few hours? Explain why. (pg. 159)

The bacteria in a hamburger might grow rapidly because they have ideal conditions to reproduce: optimal temperature, moisture, and a food source.

10. List 3 types of communicable disease. (pg. 163)

Communicable diseases are caused by pathogens. They spread from organism to organism.

Look for three of the following diseases: common cold, flu, anthrax, dysentery, E. coli 0157, Giardia, botulism, salmonellosis, meningitis, STDs, chicken pox, typhoid fever, smallpox, cholera, plague, Ebola.

11. List 3 types of non-communicable disease. (pg. 163)

Non-communicable diseases are not caused by pathogens. They cannot be passed directly from one organism to another.

Look for three of the following diseases: diabetes, cancer, arthritis, depression, stress, chronic fatigue, emphysema, heart disease, obesity, cystic fibrosis, malnutrition.

12. How does a clean environment protect you from disease? (pg. 164)

**A clean environment reduces the number of viruses and bacteria.
If there are fewer pathogens, you are less likely to get sick.**

13. List four factors that affect the risk of developing a non-communicable disease. (pg.165)

Many non-communicable diseases are the result of bad environmental conditions, poor diets, unhealthy lifestyles, and genetics.

14. What is poverty and how does it contribute to the spread of disease? (pg. 166)

Poverty means that basic needs for food, water, health care, and education are not met.

People living in poverty have an increased risk of getting a communicable disease because of improper diet, contaminated water, and lack of sanitation and personal hygiene.

KEY

15. What types of disease kills most people in poor countries? In rich countries?
Explain why you think this is so. (pg. 167)

Communicable diseases kill most people in poor countries.

Non-communicable diseases kill most people in rich countries.

- **In poor countries, communicable diseases kill people because they live in unsanitary conditions and do not have access to health care.**
- **In rich countries, communicable diseases are more easily controlled because people live in sanitary conditions and have access to health care and education.**
- **In rich countries, non-communicable diseases are more likely to kill people because of lifestyle factors such as lack of physical activity, diets high in fats and sugars, use of drugs and alcohol, stress, and smoking.**

16. Explain how each of the following methods of food preservation protects food from pathogens: drying, canning, salting, smoking, refrigeration, freezing, chemical additives, vacuum packing, irradiation, pasteurization. (pg. 168)

- **drying — removes moisture necessary for pathogens to grow**
- **canning — boiling food kills pathogens**
- **salting — removes moisture necessary for pathogens to grow**
- **smoking — removes moisture necessary for pathogens to grow**
- **refrigeration — slows down growth of pathogens**
- **freezing — causes pathogens to go dormant, thus preventing their growth**
- **chemical additives — slows down growth of bacteria**
- **vacuum packaging — removes oxygen needed for the growth of most bacteria**
- **irradiation — kills pathogens**
- **pasteurization — kills bacteria**

17. List 5 recommendations for protecting food preparation surfaces and utensils from contamination. (pg. 169)

- **Wash work surfaces and utensils thoroughly with soap and hot water.**
- **Sanitize contaminated surfaces with a chlorine solution.**
Follow the instructions on the chlorine product.
- **Wash hands with soap and hot water before and after handling raw meat, seafood, or eggs.**
- **Wear protective clothing (e.g., rubber gloves, clean apron, hair net) when preparing and handling food.**
- **Sanitize sponges and cloths with a chlorine solution.**

KEY

18. Explain why you would put an antiseptic and a disinfectant in a first aid kit. (pg.170)

- **Antiseptic — disinfects skin to protect an open cut or wound from bacteria**
- **Disinfectant — sterilizes objects such as needles or scissors used to treat an open cut or wound**

19. What is the purpose of sterilization? (pg. 170)

Sterilization kills pathogens.

20. List 4 aseptic practices. (pg. 170)

- **using sterile bandages, instruments, linens, masks, and clothing during surgery**
- **scrubbing all exposed skin of the surgeon before and after surgery**
- **having all surgical staff wear rubber gloves to prevent the spread of pathogens that might hide under fingernails.**

In addition, antiseptics are used on people to disinfect the skin.

Such antiseptics include iodine solution and alcohol.

Disinfectants are used on objects and surfaces to sterilize them.

Some common disinfectants include iodine, chlorine, bleach, and antibacterial soaps.

KEY

Chapter 10: Society's Response to Disease

Define the following:

Pandemics

diseases that affect a large proportion of the population in many countries; farther-reaching than epidemics (p. 176)

Epidemics

diseases that spread through the population faster and more widely than would normally be expected; not as widespread as pandemics (p. 176)

Quarantine

isolate sick people, often in special facilities; place where such people are isolated (page 180)

Public Health

government and other agencies working together to develop and maintain healthy communities (p. 180)

Vaccinations

injection of vaccine; an antigen; see vaccine (pp. 182, 201)

Migration (pg. 184)

diseases may be carried from country to country or from rural communities to cities as people move about

AIDS (pg. 179)

a viral disease that weakens the ability of the body to fight diseases

Tuberculosis (pg. 178)

bacterial disease that damages the lungs.

Bubonic Plague (pg. 178)

The bubonic plague is a dangerous bacterial disease with symptoms of fever, chills, and swollen glands.

Advisory (pg. 185)

when there is a major health concern, health authorities release warnings or advisories to alert you.

These include: food recalls, boil-water advisories, mass vaccinations, road safety

KEY

1. Who is responsible for the preserving the quality of water you use and drink? (pg. 174)
everyone has a certain amount of responsibility for keeping water safe to drink.
2. Why was smallpox tuberculosis and the flu so deadly to First Nations Peoples? (pg. 176)
They were not able to fight the diseases because they had never before been exposed
3. What were the effects of disease on Aboriginal Groups? (pg. 176)
The result of so many deaths was tragic.
 - **With fewer hunters, more people starved.**
 - **Many young women died. Birth rate in Aboriginal communities slowed.**
 - **Many elders died before passing on knowledge of important traditions.****The knowledge of some customs was lost.**
4. How is a pandemic different from other diseases? Give an example of a pandemic disease. (pg. 176)
Pandemic diseases affect a large proportion of the population over a large geographic area and are difficult to control.
examples: AIDS, bubonic plague, Spanish flu.
5. When does an epidemic become a pandemic? (pg. 178)
An epidemic becomes a pandemic when the disease spreads uncontrollably over a large geographic area.

6. Complete the following table:

	Bubonic Plague	Tuberculosis	Spanish Flu	AIDS
Social Conditions	– overcrowded cities – limited clean water – no waste disposal facilities – poor sanitation and hygiene – high population of rats that carried the disease	• Aboriginals had no previous exposure to the disease • overcrowding in boarding schools • poor air circulation • lack of sanitation • poor diets	– soldiers travelling during WWI spread flu to others with no previous exposure – poor sanitation – poor diets – little or no access to medical care because health-care professionals were treating soldiers	– having sex – unsafe sexual practices – sharing needles – infants born from infected mothers – transfusions of infected blood
Impact of the pandemic	• 30-50% of the population died • over 100 million deaths worldwide • People with the plague were put into quarantine. They were held in a sealed-off area until they died or got better and no longer posed a disease threat to others. Only authorities with special permission could enter the quarantine area. • crime rate dropped	• patients were sent to special treatment facilities called sanatoriums • many children were put up for adoption instead of being returned to their families • Aboriginal adults lost contact with their communities	• in Canada, more people died from Spanish flu (50 000) than from the war • 21 million deaths worldwide • loss of income for businesses such as theatres and restaurants because of fear of catching the flu • crime rate dropped	• fourth leading cause of death worldwide • high costs for treatment; no cure • fear and confusion about AIDS results in negative treatment of infected individuals
When and where the disease was present	- From 1346 to 1350, it spread through Asia and Europe. - There were two more pandemics — one in 1542, the other in 1900.	- It was common in Europe and spread by European sailors and explorers. - In Canada, it was responsible for illness and deaths among Aboriginal peoples during 1950s - 1960s - More than 70% of Inuit were infected	- The Spanish flu is a viral disease that spread around the world during 1918. It killed thousands of people on every continent except Antarctica.	- AIDS was officially recognized as a disease in 1980. It is common around the world and increasing in severity.

KEY

7. Why did society begin to take action to protect the health of the community? (pg. 180)
People began to understand the connection between sanitation and disease. Officials ordered streets cleaned and quarantined sick people in special facilities.
8. What jobs are the provincial health department responsible for? (pg. 181)
jobs as:
- **monitoring the safety of drinking water,**
 - **checking the pollutants in the air we breathe, and**
 - **taking measures to avoid outbreaks of disease.**
9. What does the Regional Health Authority do when it comes to water quality? (pg. 181)
- **Water is tested on a regular basis to ensure it is safe for drinking, cooking, bathing, or swimming.**
 - **Water is treated to make it safe to drink.**
- Treatment plants use chemicals such as chlorine to kill harmful bacteria.**
10. How do landfills prevent toxic chemicals from entering the water system? (pg. 181)
Before the landfill is used, the ground is sealed to prevent toxic chemicals from accidentally entering the water supply.
11. Why do we treat human and animal waste? (pg. 182)
To make water safe to drink
12. If the air quality is between 50 to 100 (Poor to Very Poor) what kind of alert or warning do you think would be sent out? (pg. 182)
people are advised to stay indoors and avoid heavy physical exercise.
13. List 6 vaccinations that are given by the Health Authority. (pg. 182)
In Canada, children are often vaccinated against diphtheria, whooping cough, measles, mumps, tetanus, polio, and rubella.
People who travel to other countries may get shots against yellow fever, typhoid fever, or hepatitis.
14. List the 4 points put forward by the BAC campaign. (pg. 183)
- **Clean — wash hands and surfaces often**
 - **Cook — cook to proper temperature**
 - **Separate — do not cross-contaminate**
(ex. keep separate cutting boards for meats and vegetables)
 - **Chill — refrigerate promptly**

15. Complete the following table (pg. 184):

<u>Factor</u>	<u>Concern</u>	<u>Ways to reduce risk</u>
Population growth and overcrowding	• overcrowded conditions increase the chances of being exposed to pathogens	• disease monitoring • Emergency response plans to deal with a community with disease
Migration	Diseases may be carried from country to country or from rural to urban	• Disease monitoring keeps an eye on the level of disease. • When levels are high, emergency response plans may prevent migration. • Travelers and their belongings are disinfected to prevent the spread of pathogens.
lifestyle	People eat out in restaurants	• Restaurant inspections • sanitary workplaces
Travel	• increased exposure to disease (for example, eating improperly prepared foods, using unsanitary public washrooms)	• Travellers receive special vaccinations before travelling to some countries. • Travellers receive brochures explaining how to protect themselves from pathogens in food and water.
Food supply	• food comes from many countries • food may harbour pathogens	• Food Inspectors check food supply • Transportation vehicles are refrigerated
Sexual activity and childbirth	• sexual contact results in spread of sexually transmitted diseases (STDs) • underweight babies • unhealthy mothers	• STD awareness programs provide the facts. • Prenatal nutrition programs teach mothers proper nutrition. • Awareness education programs for fetal alcohol spectrum disorder reduce the number of mothers who drink during pregnancy.

KEY

16. Do you think it is important to have public health guidelines? Use examples to help you explain why or why not. (pg. 187)

Yes, it is important to have public health guidelines to monitor public safety.

- **Good personal hygiene and treating and disposing of wastes properly reduce the spread of pathogens.**
- **Disease monitoring keeps an eye on the level of disease.**
- **Emergency response plans help a community deal effectively with an outbreak of disease.**
- **Restaurant inspections ensure that food-handling practices are safe and eating areas and utensils are kept sanitary.**
- **Vaccinations and health information can help people protect their health when they travel.**
- **Food inspectors test food for pathogens.**
- **AIDS- and other STD-awareness programs provide people with the facts they need to protect themselves.**

17. Read the Walkerton's Water Tragedy Timeline on page 186.

Do you think that the PUC manager should face criminal charges?

Explain why you think the way you do.

If you think that he should be charged what do you think he should be charged with?

ANSWERS WILL VARY

Chapter 11: Protecting Yourself from Disease

Define the following:

Antibodies

protein compounds released by white blood cells to react with and destroy antigens (p. 197)

Physical defenses

body's physical barriers against pathogens; include the skin and secretions such as mucus (p. 194)

Inflammatory response

Tissue swelling and increased blood flow; the body's second line of defense against further damage when the physical barrier of the skin is breached (p. 195)

Immune System

body's third line of defence against disease; targets specific pathogens (p. 195)

Macrophages

specialized white blood cells of the immune system that surround and destroy damaged cells, foreign protein, and pathogens (p. 195)

Antigens

structures attached to all cells; they help the body's immune system identify pathogens (p. 197)

Inherited Immunity

resistance that organisms have to most diseases; passed on genetically (p. 199)

Acquired Immunity

body's immunity to pathogens developed throughout life by natural exposure or vaccination; see active immunity and passive immunity (p. 199)

Active Immunity

**body's process of developing antibodies against certain diseases;
see acquired immunity (p. 199)**

Passive Immunity

**Antibodies introduced into the body to create immunity;
see also acquired immunity (p. 199)**

Vaccine

antigen injected into the body as a catalyst for the body to begin producing antibodies; see also vaccination and immunization (p. 201)

Immunization

catalyst for the body to begin producing antibodies; see vaccination (p. 201)

Over-the-Counter Drugs

Medications sold without a doctor's prescription (p. 204)

Antibiotic

drug that fights bacterial infections by either killing bacteria or preventing them from reproducing (p. 204)

KEY

1. Where does penicillin come from? (pg. 192)

Penicillin, the first antibiotic developed, comes from mold, which grows on bread!

2. What is the first line of defense against infection? (pg. 194)

Physical defenses

3. Complete the chart on Physical Defenses (pg. 194):

<u>Structure</u>	<u>Function</u>
Nose & Sinuses	• Contains hair and mucous, which trap pathogens before they can enter the respiratory system. • Warm incoming air.
Tonsils and Adenoids	• Catch pathogens before they enter the body. May become infected to help the immune system resist and fight future infections. • Store white blood cells.
Eyelids	• cover the eyeball to keep out pathogens, dust, and other particles • blink automatically to repel objects
Tears	• consist of salty fluids that wash eyes clean
Eyelashes	Protect the eye.
Hair	• traps or repels pathogens
Ear	Contains wax which traps pathogens and particles.
Skin	Consists of a thick barrier with an acidic surface that prevents pathogens from entering. Prevents loss of body fluids.
Stomach	• contains hydrochloric acid that kills pathogens

4. Why do smokers have a nagging cough? (pg. 194)

Cigarette smoke can disable the tiny, hair-like projections called cilia that line the air passages and trap pathogens. Without cilia, more pathogens reach the lungs. The nagging cough that smokers get is the only way for the lungs to remove debris from toxins in cigarettes.

5. What is the 2nd line of defense? (pg. 195)

Inflammatory response

6. List 3 different things that can trigger an inflammatory response. (pg. 195)

- **If pathogens get past the physical barrier created by the skin, the body sends white blood cells, that are apart of the immune system, to repel the invader.**
 - **An inflammatory response can also occur when the skin is hit, for instance, when you fall and hit the ground. This response is caused because blood oozes from the ruptured blood vessels into surrounding tissue. White blood cells engulf damaged cells and cell parts at the site of the bruise in order to clean up and heal the bruised area.**
 - **Any kind of infection (e.g. tonsillitis or appendicitis) will trigger an inflammatory response.**
- The 'itis' ending of these words means "inflammation of."**

7. What is happening when your skin swells, turns red, and gets warm around a sliver? (pg. 195)

Inflammatory response

8. What do macrophages destroy? (pg. 195)

destroy damaged cells and pathogens such as viruses and bacteria

9. What is the 3rd line of defense? (pg. 195)

the formation of antibodies in the immune system.

10. Read the following scenarios and write whether the following have antigens. (pg. 197)

- A virus enters the blood stream. **YES**
- A red blood cell travels from the heart to the lungs. **YES**
- Bacteria transferred from a piece of broken glass enters the body. **YES**
- Cancer cells in the body replicate. **YES**
- A macrophage consumes a damaged cell. **YES**

11. What does your body release when it senses an antigen? (pg. 197)

White blood cells make and release Y-shaped proteins called antibodies.

12. How do antibodies work to give us immunity? (pg. 197)

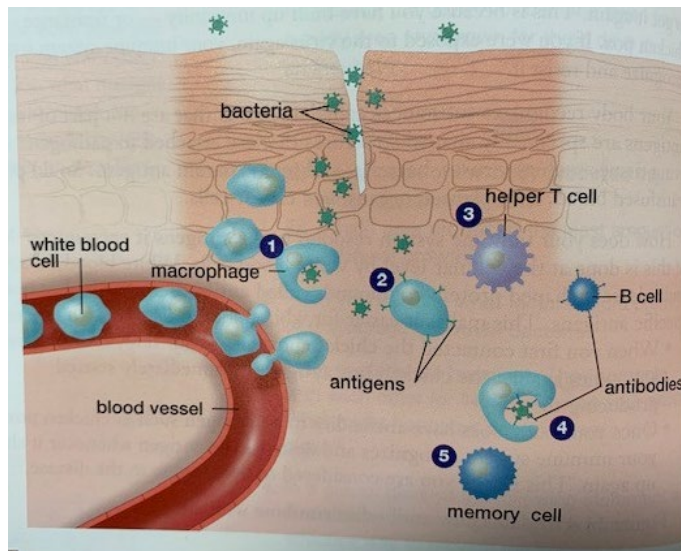
Antibodies bind to specific antigens.

This makes it easier for white blood cells to destroy them.

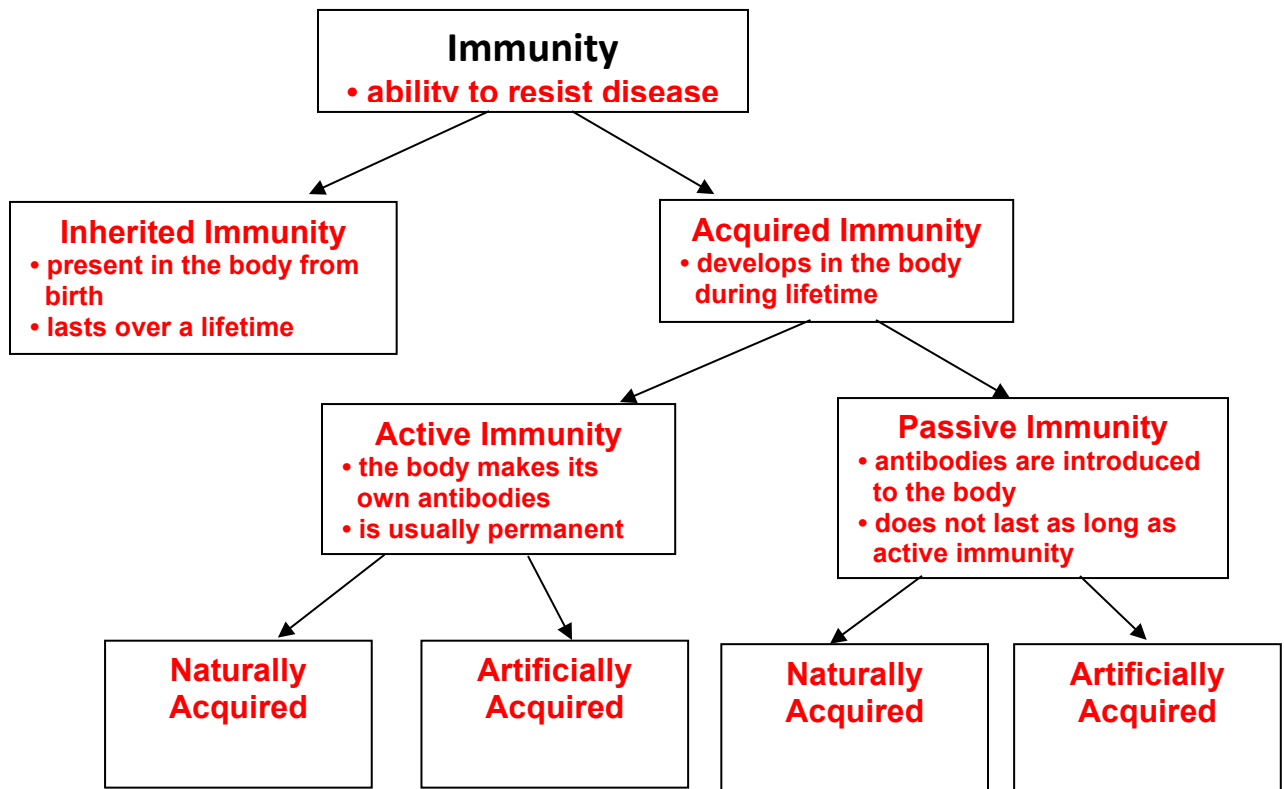
13. Look at the following diagram.

Order the following in proper chronological order from 1 to 5. (pg. 198)

- a. Antigens from dead bacteria stick out of the macrophage. 2
- b. Antibodies produced by B-cells attach to antigens and killer T-cells destroy them. 4
- c. Bacteria enter the body and macrophages surround them. 1
- d. B cells produce memory cells to store the information about how to kill the antigen so they can make the antibodies faster the next time. 5
- e. Helper T-cells read antigens and B-cells make and release antibodies. 3



14. Complete the following flowchart. (pg. 199)



15. Classify the following as a type of immunity. (pg. 199)

a. A baby gets immunity from diseases from a mother's milk.

Naturally Acquired Passive Immunity

b. You are vaccinated against smallpox.

Artificially Acquired Active Immunity

c. A cow is injected with snake venom and the antitoxin is given to you.

Artificially Acquired Passive Immunity

d. You get chicken-pox and you never get it again.

Naturally Acquired Active Immunity

16. What is the difference between passive and active immunity?

Give an example of each. (pg. 199)

• **Active immunity**
– the body produces its own antibodies
– is usually permanent

• **Passive immunity**
– antibodies are introduced to the body
– lasts only a short time

KEY

17. Why do people with A blood reject transplants from people with B blood? (pg. 200)

- **blood plasma with A-Antibodies reacts with B-Antigens**
- **causes the blood cells to clump which can lead to heart attacks and strokes**

18. Why are people with O negative blood considered universal donors? (pg. 200)

- **O negative blood has no antigens and has both A and B Antibodies**

19. What is the difference between a vaccine and an antitoxin? (pg. 201)

- **Vaccines consists of weakened or dead disease pathogens.**
After your body produces antibodies, you have lasting protection against the real disease.
- **Antitoxins are drawn from animal blood**
Antitoxins neutralize the effect of toxins and act faster than vaccines.

20. How do vaccines work? (pg. 201)

- **A vaccine contains weakened or dead pathogens.**
- **Once injected into the body, the vaccine triggers the body to begin producing antibodies against the antigens in the vaccine.**
- **After the body produces antibodies, an individual has lasting protection against a specific disease.**

21. Why should you get immunizations? Do you think some countries are justified in not giving them? (pg. 202)

- Individuals should be immunized to prevent or reduce the chances of contracting certain communicable diseases.**

22. How is your immune system related to allergies? (pg. 202)

- **An allergy develops when the immune system overreacts and becomes sensitive to a specific substance.**
- **The immune system treats the substance like a pathogen and begins building antibodies against it.**
- **The antibodies result in a variety of symptoms (e.g., runny nose) called an allergic reaction.**

23. What is the difference between over-the-counter drugs and antibiotics? (pg. 204)

Over-the-counter drugs are sold without a doctor's prescription and are used to relieve or treat symptoms such as fevers and headaches, but do not kill the viruses.

Antibiotics fight bacterial infections by either killing bacteria or preventing them from reproducing.

Each antibiotic is effective against specific types of bacteria.

They do not, however, help with a viral infection.

24. Why do antibiotics not work on colds? (pg. 204)

Colds are caused by viruses.

Antibiotics do not work against viruses.

25. List 3 problems with antibiotics. (pg. 205)

- **They kill harmful and helpful bacteria — Your body contains many helpful bacteria that aid in such things as digestion.**
- **They can cause side effects — Some people develop allergies to antibiotics.**
- **Overuse can help the development of resistant bacteria**

26. List 3 things you should do to use antibiotic safely? (pg. 206)

When using an antibiotic:

- **ask questions about precautions to take and about any side effects;**
- **read labels to find out how and when to take the medicine; and**
- **take drugs properly until you have finished the medication.**

27. What should you do when choosing Over-The-Counter Drugs? (pg. 206)

When you have a choice among several medications, read the labels carefully.

- **Find out what symptoms the medicine treats.**
Look for something that relieves those symptoms.
- **Read the warnings and list of side effects.**
- **Use exactly the amount of medication prescribed, and use it for the recommended length of time.**

Chapter 12: Genetics and Health

Define the following:

Genetics

scientific study of how characteristics or traits are passed from parents to offspring (p. 212)

Chromosomes

found in the nucleus of a cell; they contain DNA (p. 212)

DNA

complex molecule that contains genetic information; appears as long strands in the nucleus of each cell (p. 212)

Genes

basic building blocks of inherited traits that form segments that make up DNA; see also *DNA* (p. 212)

Heredity

passing of characteristics from parents to offspring (p. 216)

Dominant

inherited traits/genes that appear in offspring; tend to cover recessive traits; see also *recessive* (p. 216)

Hybrid

organism with both the dominant and recessive gene for a trait (p. 216)

Purebred

organism with identical genes for each trait; for example, the same two genes for tallness — TT or tt (p. 216)

KEY

Recessive

inherited traits/genes that do not appear in offspring except when both parents possess the trait; are covered by dominant traits; see also *dominant* (p. 216)

Genotype

letter code for the combination of genes that an organism has for each trait (p. 217)

Punnett Square

tool used by geneticists to predict the probability of offspring inheriting certain traits (p. 217)

Pedigree

family tree; diagram used by geneticists to show the history of a trait passed from generation to generation (p. 218)

Mutation

change in genetic information because of an abnormal gene (change may be beneficial, harmful, or neither) (p. 220)

Mutagens

environmental factors (anything outside the body) that cause mutation (p. 220)

Genetic Disorders

diseases or illnesses caused by alterations to genes that affect their normal functioning (p. 220)

Genetic Engineering

Transferring genes from one organism or type of organism to another (p. 223)

Amniocentesis

procedure used to determine the chromosomal makeup of a fetus and to check for abnormalities (p. 223)

KEY

1. Where in the cell are the chromosomes? (pg. 212)

In the nucleus

2. If one side of the DNA molecule looks like this A-G-C-T-G. What must the other side look like? (pg. 212)

T-C-G-A-C

3. How many chromosomes in a normal human? (pg. 213)

46 chromosomes – 23 pairs

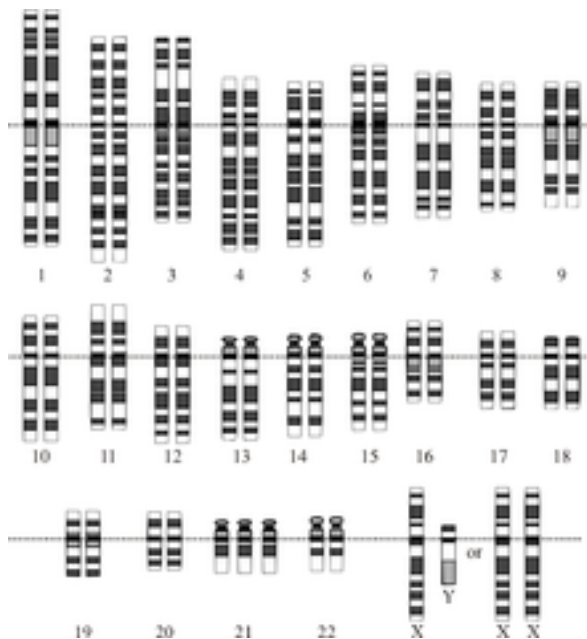
4. Here are 3 genetic disorders Identify what the person has using the following karyotypes.

Downsyndrome = Trisomy 21 (3 chromosomes), Turners Syndrome= a single X,
Kleinfelters Syndrome = XXY. (pg. 213)



Disorder:

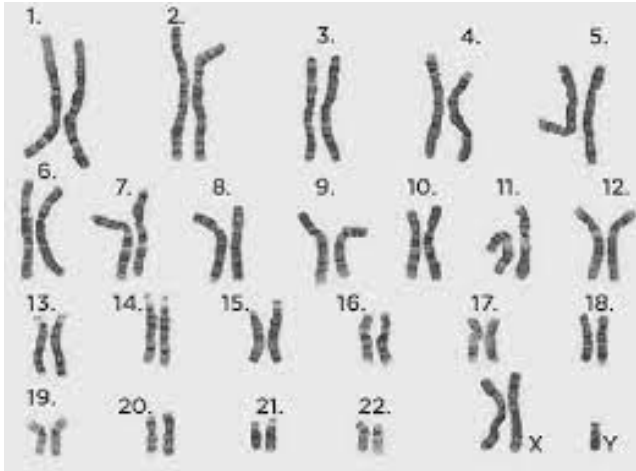
Turners Syndrome = a single X



Disorder:

Downsyndrome = Trisomy 21
(3 chromosomes)

KEY



Disorder:

Klinefelters Syndrome = XXY

5. For the following state whether the person is male or female. (pg.213)

A)XX **FEMALE**

B)XY **MALE**

6. Why can it be said that the male determines the sex of the baby? (pg. 213)

The Y chromosome is found only in males.

Females have XX chromosomes. Males have XY chromosomes.

If the male contributes an X then baby is female.

If the male contributes a Y then baby is male.

7. Who studied pea plants and became known as the father of genetics? (pg. 216)

Gregor Mendel, an Austrian monk

Use the following information to answer the next two questions.

TT or tt = Homozygous (two of the SAME genes)

Tt = Heterozygous (two different genes)

8. Make a Punnett Square using the following data. Tall Plants are Dominant to Short Plants. A heterozygous tall plant mates with a short plant. (pg. 217)

	t	t
T	Tt	Tt
t	tt	tt

9. Brown Eyes are Dominant to Blue Eyes. If a father who is homozygous for Brown Eyes mates with a woman with Blue Eyes what chance does the offspring have of having blue eyes? (pg. 217)

B – brown
b – blue

	b	b
B	Bb	Bb
B	Bb	Bb

0% chance of blue eyes

KEY

10. Draw the punnett square for the following situation and fill out the chart below.
In sheep black wool is recessive to white. So two white heterozygous sheep mate.
(pg. 217)

	W	w	
W	WW	Ww	
w	Ww	ww	

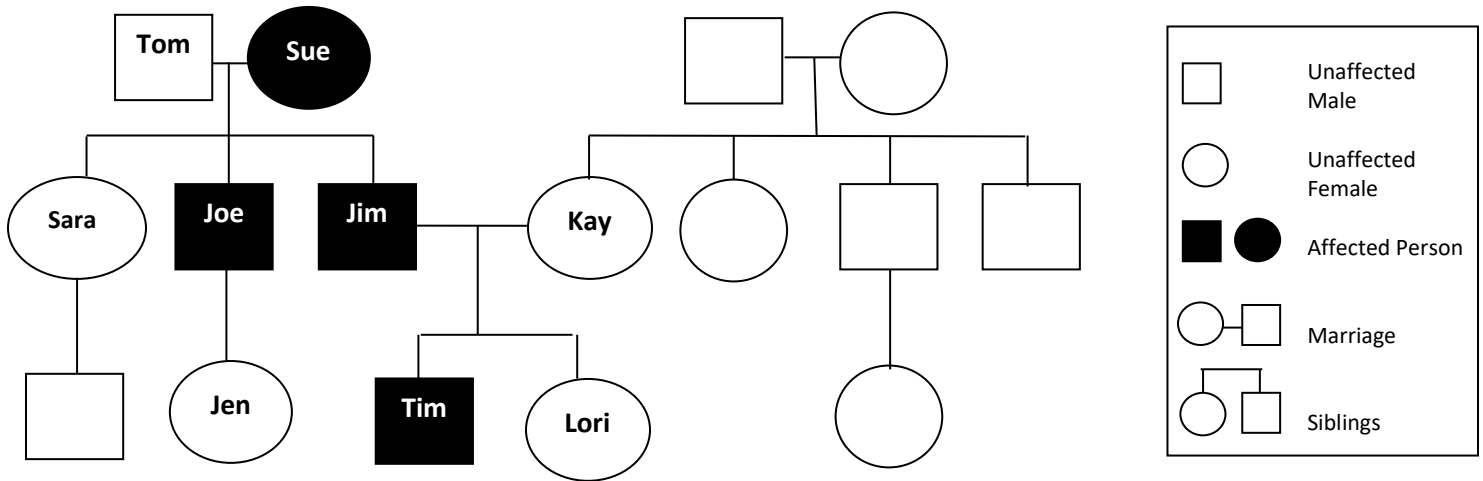
W - white

w – black

	Genotype (The Genes)	Phenotype (What it looks like)	Percentage
Offspring 1	WW	white	25%
Offspring 2	Ww	white	25%
Offspring 3	Ww	white	25%
Offspring 4	ww	black	25%

What is the ratio of White sheep to Black Sheep? **3:1**

11. Look at the following pedigree. And answer the following questions. Pg 218



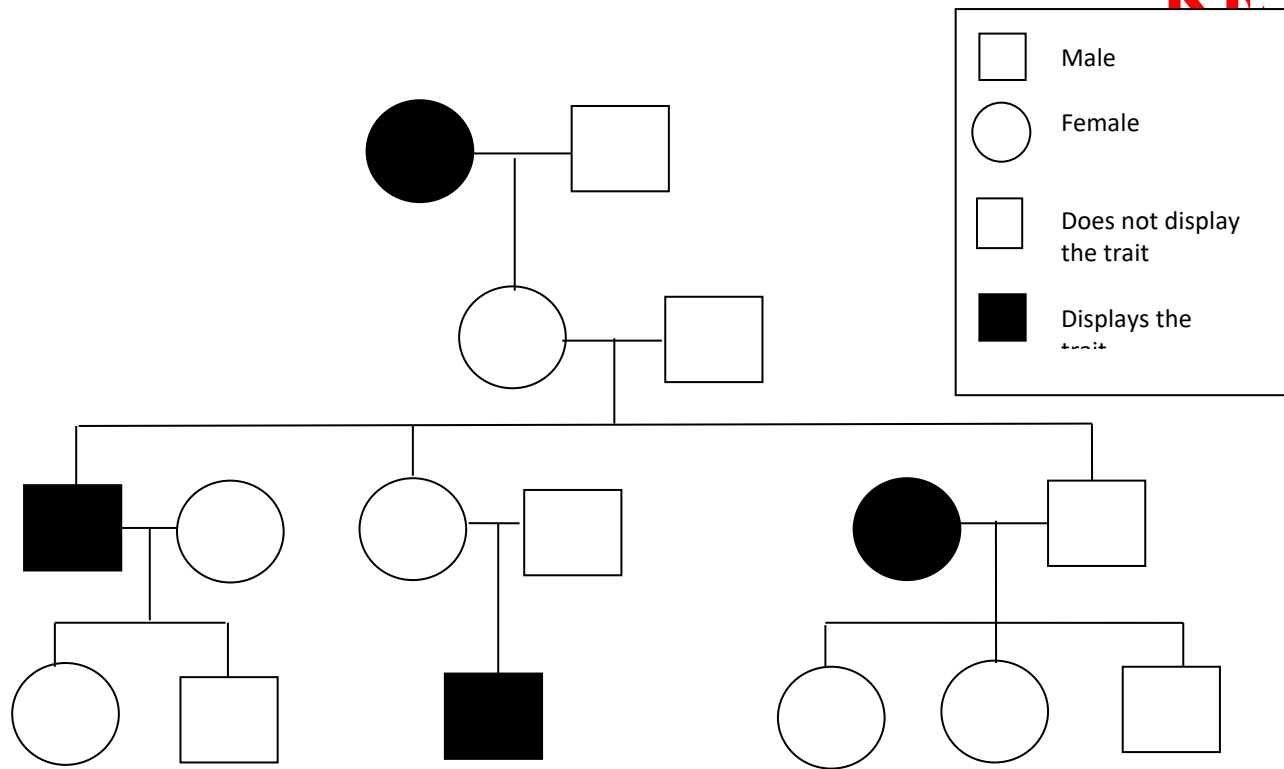
a) How many generations does this pedigree go? **3**

b) Name the infected people.

Sue, Joe, Jim, Tim

c) How many children does Jim have? **2**

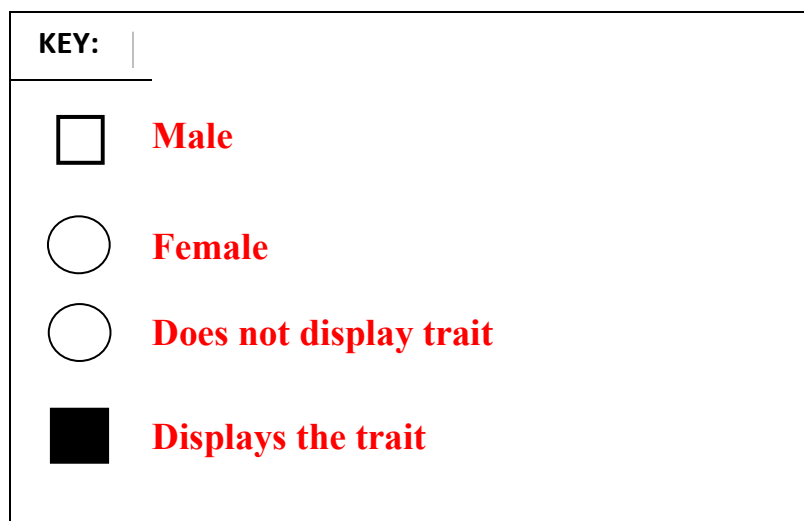
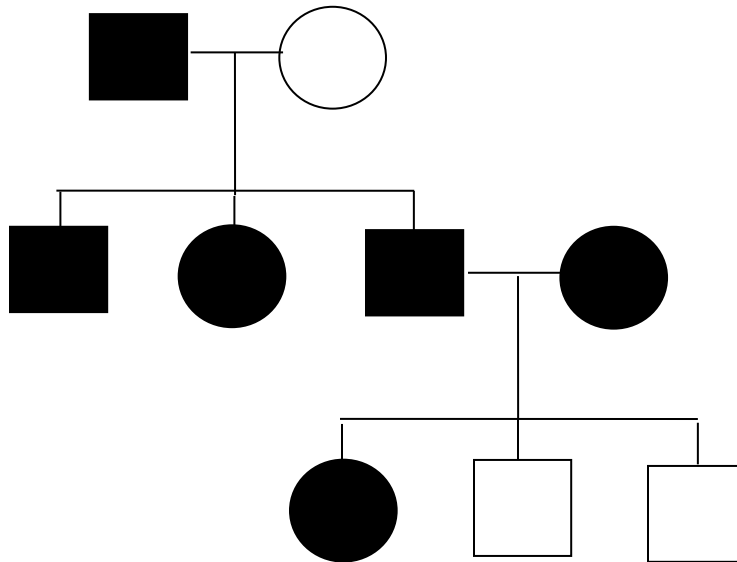
KEY



12. Using the pedigree above determine if the infected trait is dominant or recessive.
EXPLAIN why. (pg. 218)

Recessive – Two unaffected parents have affected children.
– The unaffected parents are carriers of the recessive genes.

13. Construct a pedigree using the following information.
- Brown eyes is dominant to blue eyes
 - Mr. Pederson is homozygous for brown eyes marries a Blue-eyed Girl
 - They have 3 children, 2 sons and a daughter. They all have brown eyes.
 - One of the sons and the daughter do not have children.
 - 1 son marries a brown eyed girl. They have 2 sons and daughter.
Both sons have blue eyes.
 - Be sure to include a key. (pg. 218)



14. Describe how a mutation might occur when chromosomes are copied. (pg. 220)

A change in one of the base pairs of a gene results in a defective gene

15. a) What are mutagens? (pg. 220)

Mutagens are environmental factors that cause mutation.

These include radiation; ultraviolet light; and many chemicals, including PCBs, those in cigarettes, and alcohol and other drugs.

b) Name one example of a mutagen and a genetic disorder that results from it. Pg 222

- **radiation from X-ray machines — cancer**
- **ultraviolet light in the Sun or tanning lamps — cancer**
- **exposure to alcohol during pregnancy — fetal alcohol spectrum disorder (FASD)**
- **PCBs — liver and kidney cancer**

16. Why should pregnant women not drink or do drugs? Pg 222

Pregnant women should avoid alcohol and drugs (except for prescription drugs that are recommended by their doctor) because they may harm the fetus.

17. Why might amniocentesis pose an ethical dilemma for some people? (pg. 223)

Answers may vary.

- **People who oppose abortion believe that amniocentesis tests should not be available because some parents will use the test to decide whether or not to terminate a pregnancy.**
- **People who support reproduction rights believe that amniocentesis tests should be available because parents have the right to terminate a pregnancy.**

18. Complete the following chart on advances in genetic research. (pgs. 223 and 225)

Pros of Genetic Research	Cons of Genetic Research
• Genetic disorders can be identified. • New drugs and treatments can be developed. • Genetic information helps people make lifestyle choices. • Health-care providers are obliged to respond to the health needs of their patients.	• Scientists do not know all the risks involved with genetic research. • There are ethical issues about using human embryos to harvest stem cells. • Health insurance companies might refuse to provide health insurance to some individuals. or An employer might fire someone if they had access to genetic data. • Knowledge about genetic make-up might cause an individual to adopt high-risk behaviours or cause depression. • Techniques can be misused. Genes might be altered for reasons other than curing a disease.

19. Should they allow cloning of people so people may harvest organs from their clone in the case of a medical emergency?

ANSWERS WILL VARY